Description of Course CSE 220

PART A: General Information

1	**Course Title**	: SIGNALS AND LINEAR SYSTEMS SESSIONAL
2	**Type of Course**	: SESSIONAL
3	**Offered to**	: DEPARTMENT OF CSE
4	**Pre-requisite Course(s)**	: None

PART B: Course Details

1. Course Content (As approved by the Academic Council)

Sessional based on CSE 219.

2. Course Objectives

The students are expected to:

i. Learn continuous and discrete-time signals, linear systems and their properties.
ii. Learn Fourier, DFT, Laplace and Z transforms
iii. Apply the principles of Fourier DFT, Laplace and Z transforms for analyzing signals and linear systems.

3. Knowledge required

Technical

- Basic of calculus and differential equations

4. Course Outcomes (COs)

CO No.	CO Statement After undergoing this course, students should be able to:	Corresponding PO(s)*	Domains and Taxonomy level(s)**	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
CO1	**Understand** continuous and discrete-time signals, linear systems, their properties, and the sampling process	-	C2	Lecture, Demonstration, and hands-on	Assignments and Quiz
CO2	**Apply** mathematical concepts and tools for analysis of signals and linear systems	PO1, PO5	C3	Lecture, Demonstration, and hands-on	Assignments and Quiz
CO3	**Analyze** continuous and discrete-time signals, and linear systems in time and frequency domains	PO1, PO2	C4	Lecture, Demonstration, and hands-on	Assignments and Quiz

***Program Outcomes (POs)**

PO1: Engineering knowledge; PO2: Problem analysis; PO3: Design/development of solutions; PO4: Investigation; PO5: Modern tool usage; PO6: The engineer and society; PO7: Environment and sustainability; PO8: Ethics; PO9: Individual work and teamwork; PO10: Communication; PO11: Project management and finance; PO12: Life-long learning.

****Domains**

C-Cognitive: C1: Knowledge; C2: Comprehension; C3: Application; C4: Analysis; C5: Synthesis; C6: Evaluation
A-Affective: A1: Receiving; A2: Responding; A3: Valuing; A4: Organizing; A5: Characterizing
P-Psychomotor: P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Organization

5. Mapping of Knowledge Profile, Complex Engineering Problem Solving and Complex Engineering Activities

COs	K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
CO1		√	√	√	√	√			√	√	√	√				√			√	
CO2		√	√	√	√	√			√	√	√	√				√	√			
CO3			√	√	√	√		√	√	√	√				√	√		√		√

K-Knowledge Profile:

K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline; **K2:** Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline; **K3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline; **K4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline; **K5:** Knowledge that supports engineering design in a practice area; **K6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline; **K7:** Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability; **K8:** Engagement with selected knowledge in the research literature of the discipline

P-Range of Complex Engineering Problem Solving:

P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach; **P2:** Involve wide-ranging or conflicting technical, engineering and other issues; **P3:** Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models; **P4:** Involve infrequently encountered issues; **P5:** Are outside problems encompassed by standards and codes of practice for professional engineering; **P6:** Involve diverse groups of stakeholders with widely varying needs; **P7:** Are high level problems including many component parts or sub-problems

A-Range of Complex Engineering Activities:

A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies); **A2:** Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues; **A3:** Involve creative use of engineering principles and research-based knowledge in novel ways; **A4:** Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation; **A5:** Can extend beyond previous experiences by applying principles-based approaches

6. Lecture/ Activity Plan (Tentative)

Week	Lecture Topics	Corresponding CO(s)
Week 1	Lecture on Python (Introduction, Installation, Tools, Basic Python: variables, user input, branching, loops, functions etc.)	CO2
Week 2	Lecture on Python (Numpy, matplotlib, plotting basic signals)	CO2
Week 3	Practice class on Python	CO2
Week 4	Online on Signals and their properties etc.	CO1 and CO2
Week 5	Offline on Convolution Declaration + Practice class on Convolution	CO1 and CO2
Week 6	Offline Evaluation (on Convolution)	CO1 and CO2
Week 7	Online on Convolution + Offline declaration on Fourier Series	CO1 and CO2

Week	Lecture Topics	Corresponding CO(s)
Week 8	Offline Evaluation on Fourier Series Offline declaration + Practice class on Continuous Fourier Transform (CFT)	CO1, CO2 and CO3
Week 9	[Holiday]	CO1, CO2 and CO3
Week 10	Online + Offline evaluation on Continuous Fourier Transform (CFT) Offline declaration on DFT & FFT	CO1, CO2 and CO3
Week 11	Practice on DFT & FFT	CO1, CO2 and CO3
Week 12	Online + Offline evaluation on DFT & FFT Offline declaration on Sampling/Filtering/Laplace/Z-transform	CO1, CO2 and CO3
Week 13	Quiz	CO1, CO2 and CO3
Week 14	Online + Offline evaluation on (Sampling/Filtering/Laplace/Z-transform)	CO1, CO2 and CO3

7. Assessment Strategy

- Class Attendance: Class attendance will be recorded in every class.
- Assignments: There will be some online and offline assignments
- Quiz Exam: A comprehensive Quiz exam will be held at the end of the semester.

8. Distribution of Marks (Tentative)

Attendance:	10 %
Assignments:	60%
Quiz Exam:	30%
Total:	100%

9. Textbook/Reference

a. Oppenheim, Alan V., and A. S. Willsky. Signals and Systems.